PROJECT REFLECTION · LUCA TOMARELLI

From Notebook to Production

Pipeline, API and containerisation

i. Steps taken to complete the project

- Went through the notebook to understand how data cleaning, processing and modeling were done.
- Identified which steps needed to be replicated in the Python scripts, separating those used only for exploration from those that were actually part of the workflow.
- Refactored the selected parts into Python modules and built a complete pipeline that takes the raw data, cleans it, processes it and uses it to train the model.
- Added a `model` flag with two options, `full` and `base`: `full` is the default and trains the model on the full feature set, while `base` uses only 5 features and trains the model served by the app.
- Built a FastAPI app with two routes: a root `GET` and a prediction `POST`, which takes 5 inputs and returns the model's price prediction.
- Containerised the app with Docker.

ii. Challenges

- The notebook contained many explorations and dead ends, so the first challenge was bringing order to it.
- Writing a Dockerfile that could serve the trained model.
- By contrast, the refactoring, the pipeline and the FastAPI app went fairly smoothly.

iii. What I would do differently with more time

- Select better features for the production model to improve its score, while keeping it easy to use for whoever has to enter the input values.
- Build a small frontend to serve the app properly.